

L-PHYM: Long-Horizon Language-Driven Physics-Based Motion Control

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GitHub: <https://github.com/AesliteC/L-PHYM>

Project page: <https://aesslitec.github.io/L-PHYM/>

Abstract—Long-horizon language-driven humanoid control sits between text-to-motion generation and physics-based whole-body control. Most text-to-motion systems are trained and evaluated on short, single-action clips, whereas robot commands often describe ordered subgoals such as locomotion, turning, reaching, sitting, and gesturing within one instruction. Such commands require more than local motion realism: the generated reference must preserve action order, maintain reasonable transitions and duration, and remain suitable for a physical humanoid whose morphology, contact, balance, and actuator constraints differ from a kinematic human skeleton. L-PHYM addresses this setting with a two-part pipeline that first generates long kinematic references in MoConVQ’s discrete motion space and then evaluates their physical feasibility through H1 retargeting and teacher-level tracking. For reference generation, we replace a brittle hand-coded HumanML3D-to-MoConVQ state mapping with a BVH-native route that uses MoConVQ’s native retargeting, tokenizer, and segment-aware text-conditioned GPT fine-tuning. On a 100-prompt held-out HumanML3D-test suite, the selected model improves several long-prompt rollout and retrieval indicators over the released MoConVQ baseline, while the gains are not uniform across all approximate evaluator metrics. For humanoid feasibility, candidate motions are converted to an AMASS-like H1 fitting interface, normalized for coordinate convention and root height, diagnosed with contact and smoothness checks, and retargeted with an optimized batched pipeline. A privileged HOVER-style teacher trained with PPO is then evaluated on multiple AMASS-style H1 benchmark sets. The resulting teacher achieves strong tracking performance on favorable motion sets, showing that the retargeting and control pipeline can produce physically feasible humanoid rollouts from selected long-horizon references.

I. INTRODUCTION

Language-conditioned motion generation is often evaluated on short, single-action prompts. This setting is insufficient for robot control, where a command naturally contains ordered subgoals: walk forward, turn, reach, crouch, stand, wave, or combine locomotion and upper-body gestures. Long instructions stress a text-to-motion model in several ways. The generator must preserve the order of actions, produce a continuous reference without abrupt state changes, avoid early termination, and remain inside a motion distribution that can later be retargeted and tracked by a physical humanoid.

Kinematic motion quality is only the first part of the problem. A motion that looks plausible in a skeleton viewer can still be unsuitable for robot control because contacts,

root height, balance, and morphology constraints may be inconsistent with a target humanoid. For this reason, L-PHYM separates the problem into reference generation and physical tracking. The first stage generates long-horizon kinematic references from language in MoConVQ’s discrete motion-token space [12]. The second stage retargets the generated references to an H1 humanoid and trains a physics-based tracking policy so that the long kinematic plan can be executed under balance, contact, joint-limit, and torque constraints.

The completed work covers both stages. In Stage I, the central issue is how to turn HumanML3D compositions into MoConVQ-compatible supervision without damaging the motion-token distribution. A direct route would compose HumanML3D clips, map their 22-joint kinematic states into the MoConVQ character state, encode the resulting observations into residual-VQ tokens, and fine-tune the text-conditioned GPT. Although this route creates many training samples quickly, our experiments show that the representation interface becomes the dominant failure point: a hand-coded HumanML3D-to-MoConVQ state mapping can produce good token-accuracy curves while creating an unhealthy token cache, and the generated videos remain unstable, repetitive, or semantically weak. The final Stage I pipeline therefore keeps HumanML3D as the language-motion source but replaces hand-coded state conversion with a BVH-native route: long HumanML3D compositions are exported to BVH, passed through MoConVQ’s native character retargeting path, encoded into residual-VQ token sequences, and used to fine-tune the text-conditioned GPT. Segment metadata is preserved throughout the pipeline so that training and inference use the same multi-segment protocol rather than relying on fragile string splitting.

The second part of the report evaluates whether these kinematic references can become physically meaningful H1 motion libraries. Stage II first converts candidate motions into the AMASS-like interface expected by the H1 fitting pipeline [?], including poses, root translations, shape parameters, and frame rate. It then normalizes coordinate conventions and root height, estimates contact states, and optimizes an H1 trajectory with keypoint, contact, joint-limit, and smoothness terms. This retargeting stage is also where invalid references are diagnosed: motions with poor grounding, excessive joint velocity, or large acceleration spikes are filtered or marked as difficult before policy rollout.

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Fig. 1. Stage I fine-tuned rollout for a long HumanML3D-derived prompt. The instruction asks the character to pick up an object-like target, scrub at waist level, and then walk forward.

After retargeting, the H1 motion library is used to train and evaluate a HOVER-style privileged teacher in Isaac Lab [4, 6]. The teacher is optimized with PPO [9] to track the reference motion while satisfying balance, contact, energy, and termination constraints. In the final experiments, we evaluate the trained teacher checkpoint on multiple AMASS-style H1 benchmark sets under the same script and environment count. The results show that the current Stage II pipeline tracks favorable sets with high success, while larger or more varied sets still expose retargeting and tracking failures that should be addressed before student distillation.

The main contributions are:

- a boundary-aware construction pipeline for long HumanML3D motion-language sequences that preserves clip captions, segment boundaries, and transition metadata;
- a BVH-native HumanML3D-to-MoConVQ conversion route that avoids the token collapse observed with hand-coded state conversion and produces healthier residual-VQ supervision;
- a segment-prefix fine-tuning and inference protocol that extends MoConVQ’s text-conditioned GPT from short prompts to ordered multi-action instructions;
- an evaluation suite combining rollout diagnostics, qualitative videos, and approximate HumanML3D FID/R-precision metrics on held-out long prompts;
- a generated-reference-to-H1 execution path with BVH-to-AMASS-like conversion, optimized batched retargeting, grounding normalization, and teacher-student tracking-policy training.

II. RELATED WORK

A. Text-to-Motion Generation

HumanML3D provides paired natural-language captions and 3D human motions and is a standard benchmark for text-to-motion generation [1]. Recent text-to-motion models include diffusion-based approaches such as MDM [10] and discrete autoregressive approaches such as T2M-GPT [13]. These methods are typically trained and evaluated on single captions paired with short motion clips. Such models can produce diverse and natural motions for local actions, but long-horizon command following introduces additional requirements: action order must be preserved, segment transitions should remain smooth, and generated duration should match the number and complexity of sub-instructions.

L-PHYM differs from standard short-clip text-to-motion generation in its output interface. The generated motion is not only a visualization artifact; it is intended to become a reference for a later humanoid tracker. This motivates using a motion representation already connected to a simulated character controller rather than optimizing only for kinematic skeleton quality.

B. Discrete Motion Representations

Vector-quantized representations make continuous motion compatible with language-model-style sequence prediction [11]. MotionGPT casts motion tokens as a language-like sequence [5], and T2M-GPT shows that discrete motion tokens can be generated autoregressively from text [13]. MoConVQ extends this idea to physics-based character control by learning

a residual vector-quantized latent space connected to a motion decoder/controller [12]. This makes MoConVQ an appropriate base system for L-PHYM: a successful long-prompt generator can reuse the same token space for text-to-motion, tracking, and downstream control.

C. Physics-Based Humanoid Control

Physics-based tracking systems such as DeepMimic and AMP demonstrate that motion imitation can produce robust simulated skills [7, 8]. Recent humanoid systems such as H2O, OmniH2O, and HOVER focus on human-to-humanoid retargeting and whole-body control [3, 2, 4]. These systems motivate the planned second stage of L-PHYM, where generated long-horizon references will be filtered, retargeted, and tracked under physical constraints.

III. METHODOLOGY

L-PHYM is formulated as a two-stage language-to-control framework (Figure 2). Stage I maps an ordered natural-language instruction to a long kinematic motion in MoConVQ’s discrete token space. It constructs boundary-aware multi-segment HumanML3D compositions, passes them once through a MoConVQ-compatible kinematic-file interface to the native character retargeting and tokenization pipeline, and fine-tunes MoConVQ’s Text2Motion MoConGPT to generate residual-VQ token sequences from explicit segment metadata. Stage II will take the resulting kinematic references as the input to humanoid retargeting and physics-based tracking.

The central methodological issue is that long language instructions require both semantic sequencing and physical execution. A single text-to-motion model can learn the order and duration of sub-actions, but it should not be expected to also resolve robot morphology, balance, contacts, and actuator constraints. We therefore use Stage I to produce an ordered kinematic motion in MoConVQ’s token space, where the pretrained decoder/controller can already realize character motion, and leave robot-specific feasibility to Stage II. In this design, Stage I provides the ordered kinematic reference and token metadata, while Stage II turns that reference into a physically executable humanoid rollout.

A. Stage I: Long-Horizon Text-to-Motion Generation

1) **Boundary-Aware HumanML3D Composition:** Long training samples are built from short HumanML3D clips. Each source clip contains a motion sequence and one or more natural-language captions. A long sample is constructed by selecting an initial clip and then repeatedly selecting compatible successor clips from the same split. Before scoring a candidate transition, the candidate clip is aligned to the previous clip by root translation and yaw. The boundary score is

$$s = d_r + 0.5d_v + d_{yaw} + d_f + 0.5d_{fv}, \quad (1)$$

where d_r is the root-position gap, d_v is the root-velocity gap, d_{yaw} is the facing-direction mismatch, d_f is the foot-height gap, and d_{fv} is the foot-velocity gap. This score is not a full

physical validity test, but it removes obvious discontinuities that make retargeting and tokenization unstable.

The composition process preserves the selected source clip identifiers, the frame range of each segment, the caption attached to each segment, and the joined long caption. These fields define the supervision used later by MoConGPT: segment boundaries determine which frames belong to each local action, segment captions provide the text condition for those frames, and the joined caption remains the human-readable long instruction. Keeping this metadata also makes inference unambiguous, because segment boundaries are carried by the data rather than recovered from fragile text rules.

2) **MoConVQ-Native Retargeting and RVQ Cache Construction:** Each composed HumanML3D motion is first exported as a MoConVQ-compatible BVH file. The exporter copies the hierarchy and channel order from MoConVQ’s base character template, converts the HumanML3D joint sequence into per-joint local Euler rotations under that hierarchy, and writes the MOTION section with the root translation and local rotation channels for every frame. The frame count, frame time, and segment timing are kept consistent with the HumanML3D composition. The resulting kinematic file can be loaded by MoConVQ’s native character-processing pipeline rather than by a hand-coded state converter.

After import, MoConVQ retargets the motion to its simulated character and converts the result into character observations. These observations are then encoded by the pretrained MoConVQ motion encoder with its residual-VQ codebooks, producing motion latents and RVQ indices. The HumanML3D compositions do not directly provide training pairs in MoConVQ’s token space, so this conversion is used to build the supervised dataset for Text2Motion MoConGPT. We store it as a segment-aligned training cache: each valid window contains the T5 features of the current segment caption, the motion-latent prefix used as context, masks for supervised positions, segment metadata, and the four RVQ target indices per motion step. The cache is therefore the bridge between retargeted kinematic trajectories and the autoregressive GPT objective: it fixes the alignment between segment text, prefix motion, and target tokens before fine-tuning. Using the native retargeting and encoding path keeps these supervised tokens close to the distribution expected by the pretrained MoConVQ decoder/controller.

3) **Segment-Aligned MoConGPT Fine-Tuning and Inference:** Text2Motion MoConGPT was designed to predict motion tokens from a bounded history of previous latents and a text condition. In the released model, the effective motion context is 50 latent tokens, so a composed long sequence cannot be supervised as one full context without truncation or misalignment. A single joined caption can also blur which sub-action should supervise which frames. We therefore train on segment-aligned windows. Each window pairs the current segment caption with a previous-segment motion latent prefix. The prefix provides the local motion history needed for continuity, while the segment caption specifies the action currently being generated. The model predicts the current RVQ token

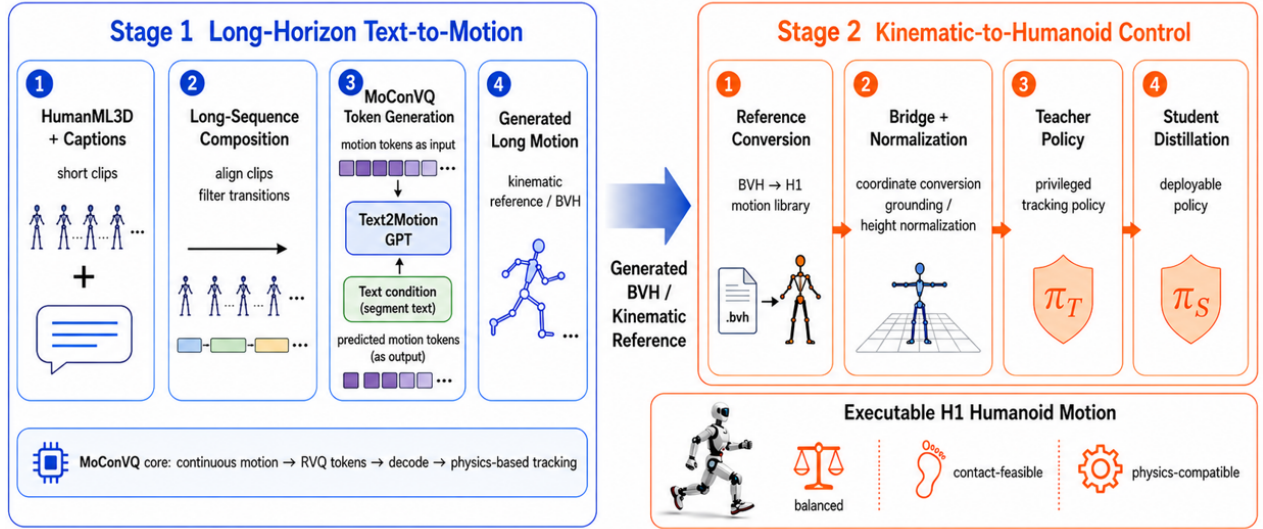


Fig. 2. L-PHYM two-stage framework. Stage I converts HumanML3D-derived long language instructions into MoConVQ motion tokens and kinematic references. Stage II retargets selected references to a target humanoid and trains a physics-based tracking policy.

tuple at each supervised time step:

$$\mathcal{L}_{CE} = - \sum_{t,d} m_t \log p_{\theta}(c_{t,d} | z_{<t}, e_{\text{text}}), \quad (2)$$

where m_t masks out prefix and padding positions, $z_{<t}$ denotes previous motion latents, and $c_{t,d}$ is the RVQ index at residual depth d .

The same segment structure is used at inference. A long prompt is stored as a joined instruction together with a list of segment strings and segment lengths. The generator iterates over the segment list, encodes the current segment text, generates its motion tokens, and carries the generated latents forward as the prefix for the next segment. The segment list is stored explicitly because temporal words are not reliable separators: in HumanML3D, a word such as “then” may appear inside a single caption describing one clip, or it may separate two intended sub-actions in a composed instruction. Explicit segment metadata removes this ambiguity.

B. Stage II: Humanoid Retargeting and Physics-Based Tracking

Stage II takes the long-horizon kinematic references produced by Stage I and turns them into physically simulated Unitree H1 tracking rollouts. The input motion is not assumed to be directly executable. A generated reference may have source-format conventions inherited from MoConVQ, residual transition artifacts from segment composition, or root and contact patterns that are plausible for a human skeleton but infeasible for H1. Stage II therefore contains two interfaces: first, a retargeting interface that converts the generated kinematic trajectory into an H1-compatible motion library; second, a physics-based tracking interface that trains and evaluates a HOVER-style teacher policy on that library [4]. Student distillation is kept as the final deployment interface, but the experiments in this report focus on teacher-level tracking.

1) Canonical Motion Interface and H1 Retargeting: The first step converts each generated sequence into a canonical human-motion interface. This pre-retargeting stage removes source-format details and standardizes units, frame rate, coordinate axes, root height, yaw convention, and left-right joint names. It also estimates contact intervals and records which body parts and time spans are semantically important for the long instruction. We write this canonical reference as

$$\mathcal{K} = \{J^{src}_{1:T}, R^{src}_{1:T}, x^{root}_{1:T}, c^{foot}_{1:T}, m^{sem}_{1:T}\}, \quad (3)$$

where J^{src} are canonical body keypoints, R^{src} are local joint rotations, x^{root} is the root trajectory, c^{foot} is the estimated foot contact state, and m^{sem} marks language-relevant body parts or time spans. This interface makes generated MoConVQ rollouts, processed BVH files, and motion-capture references comparable before robot-specific fitting.

The second step maps the canonical motion to H1. The exported reference must match the motion-library format consumed by the HOVER environment:

$$\mathcal{M}_{H1} = \{x^{root}_{1:T}, R^{root}_{1:T}, q^{H1}_{1:T}, f_{tps}\}, \quad (4)$$

where $q^{H1}_t \in \mathbb{R}^{19}$ is the H1 joint configuration. Rather than treating this as a pure skeleton-name conversion, we use a feasibility-aware fitting objective:

$$\begin{aligned} \min_{\{q_t, x_t, R_t\}_{t=1}^T} & \sum_{t=1}^T \sum_{i \in \mathcal{B}} w_i(t) \|FK_i(q_t, x_t, R_t) - J^{src}_{\rho(i),t}\|_2^2 \\ & + \lambda_c \mathcal{L}_{\text{contact}}(q_t, x_t, c^{foot}_t) + \lambda_l \mathcal{L}_{\text{limit}}(q_t) \\ & + \lambda_s \|\Delta q_t\|_2^2 + \lambda_a \|\Delta^2 q_t\|_2^2 + \lambda_r \|\Delta^2 x_t\|_2^2. \end{aligned} \quad (5)$$

Here FK_i gives the H1 forward-kinematics keypoint for body i , and $\rho(i)$ maps H1 bodies to canonical human keypoints. The contact term penalizes foot motion during contact phases,

the limit term keeps the reference inside robot joint ranges, and the smoothness terms reduce high-frequency joint and root accelerations. Motions with invalid initial root height, excessive joint velocity, excessive acceleration, or visually implausible contact are diagnosed or filtered before policy evaluation.

2) **Teacher Tracking Policy:** The H1 reference library is tracked by a privileged teacher policy following the HOVER training interface [4]. The teacher is optimized with PPO [9]:

$$\max_{\pi_T} \mathbb{E}_{\pi_T} \left[\sum_{t=0}^{H-1} \gamma^t r_t \right], \quad (6)$$

where the reward combines full-body tracking, root tracking, smoothness, energy, contact, and termination terms. The teacher receives privileged reference information and is evaluated by rolling out each reference motion from start to finish. An episode succeeds only if the policy completes the reference without losing balance, violating gravity/root-distance criteria, or making undesired contacts with non-foot body parts.

Long-horizon generated references differ from the short curated motion clips used by standard HOVER training. They contain synthetic transition boundaries, uneven segment durations, and upper-body gestures that may be rare in the training distribution. We therefore keep two adaptations in the Stage II training recipe. The first is a long-horizon window curriculum. A long reference is split into overlapping windows $\mathcal{W} = \{w_k\}$, and windows are sampled with segment and failure reweighting:

$$p(w_k) \propto 1 + \alpha n_{\text{seg}}(w_k) + \beta \mathbb{1}[w_k \in \mathcal{F}], \quad (7)$$

where $n_{\text{seg}}(w_k)$ increases coverage of rare action segments and $\mathcal{F}$ stores windows that failed during teacher evaluation. This prevents late actions and transition regions from being under-sampled.

The second adaptation is mixed-data anti-forgetting training. The teacher is trained or fine-tuned on a mixture of clean motion-capture references and generated long-horizon references:

$$\begin{aligned} \mathcal{D}_{\text{train}} &= \mathcal{D}_{\text{clean}} \cup \mathcal{D}_{\text{gen}}, \\ p(m) &= \lambda p_{\mathcal{D}_{\text{gen}}}(m) + (1 - \lambda) p_{\mathcal{D}_{\text{clean}}}(m). \end{aligned} \quad (8)$$

The clean subset preserves balance, stepping, and recovery behavior, while the generated subset adapts the teacher to the distribution of language-produced long motions. This mixture is important because a teacher that only sees noisy generated references can forget stable locomotion, while a teacher trained only on clean short clips may fail at long generated transitions.

3) **Student Deployment Interface:** Once a teacher can reliably track the mixed reference library, the same system can be distilled into a student policy for deployment. The student uses proprioceptive history and masked command features rather than the teacher’s full privileged reference observations:

$$o_t^S = [q, \dot{q}, \omega^{base}, g, a_{t-1}]_{t-L:t} \oplus (M_t \odot g_t) \oplus M_t. \quad (9)$$

TABLE I
MAIN STAGE I EXPERIMENTAL SETUP.

Item	Value
Representation route	HumanML3D compositions $\rightarrow$ MoConVQ kinematic references
Train / validation sequences	73 / 18
Train / validation windows	476 / 117
Train / validation valid RVQ tokens	85,328 / 20,756
Fine-tuning scope	base-head
Learning rate / epochs	5×10^{-6} / 3
Inference protocol	explicit segments and segment lengths

During distillation, the frozen teacher provides the target action on the same physical state,

$$\hat{a}_t = \pi_T(o_t^T), \quad a_t^S = \pi_S(o_t^S), \quad (10)$$

and the student minimizes

$$\min_{\pi_S} \mathbb{E}_t [\|a_t^S - \hat{a}_t\|_2^2]. \quad (11)$$

This component defines the intended deployment path from privileged simulation tracking to a proprioceptive policy. Since the final experiments prioritize reference feasibility and teacher tracking, student distillation is left as future work rather than reported as an experimental result.

IV. EXPERIMENTS

A. Stage I: MoConGPT Fine-Tuning

The main Stage I experiment fine-tunes the released Text2Motion MoConGPT for long-prompt generation using the selected native-retargeted HumanML3D cache. Starting from composed HumanML3D sequences, the cache provides segment-aligned supervision in MoConVQ’s token space. This setup follows the representation diagnostics in Section V, which identify native retargeting and filtering as necessary for stable MoConVQ token statistics.

Table I summarizes the training setup. The selected cache contains 73 training and 18 validation long sequences after retargeting and filtering, producing 476 training windows and 117 validation windows. The cache contains 85,328 valid training RVQ tokens and 20,756 valid validation RVQ tokens. The fine-tuning run initializes from the released MoConVQ Text2Motion MoConGPT checkpoint and uses conservative base-head tuning with learning rate 5×10^{-6} for three epochs.

The selected model optimizes consistently under the segment-aligned cache: training loss decreases from 13.8335 to 9.8848 and validation loss decreases from 14.9117 to 10.7311 (Table II). Token accuracy is reported as an optimization diagnostic rather than as the primary quality criterion: each motion step requires a tuple of four residual-VQ code indices, and a token-level match does not directly measure temporal coherence, motion coverage, or text-motion alignment. The following evaluation therefore combines rollout diagnostics with approximate text-to-motion retrieval metrics.

TABLE II
SELECTED NATIVE-RETARGETED SMALL-CACHE TRAINING CURVE.

Epoch	Train loss	Val loss	Train acc.	Val acc.
0	13.8335	14.9117	0.0579	0.0707
1	11.7993	12.5843	0.0573	0.0715
2	9.8848	10.7311	0.0598	0.0719

TABLE III
HUMANML3D-TEST LONG100 RESULTS FOR THE SELECTED STAGE I MODEL. FID AND MATCHING ARE LOWER-IS-BETTER; R@K IS HIGHER-IS-BETTER.

Metric	Baseline	Fine-tuned
Prompts	100	100
Avg. frames	1216.56	1230.96
Early-stop rate	0.44	0.41
Root path	3.3679	3.4323
Pose velocity mean	38.0968	37.5597
Approx. FID	7.0400	7.2092
Approx. R@1	0.050	0.050
Approx. R@2	0.110	0.140
Approx. R@3	0.160	0.160
Approx. matching	5.0077	4.9591

B. Stage I: Long100 Evaluation

Long100 is the main held-out evaluation suite for long-prompt generation. It is built from the HumanML3D test split by sampling 300 held-out captions with seed 13 and grouping them into 100 three-segment prompts. Each prompt stores the joined instruction, explicit segment text, and fixed segment lengths [25, 25, 25]. Long100 is a text-only held-out suite: the sampled captions are not paired with a smoothed ground-truth concatenation. It therefore evaluates generation from unseen multi-segment text, while transition smoothness is measured only through rollout diagnostics.

The released MoConGPT baseline and the fine-tuned model are compared under the same prompts, random seed, explicit segment metadata, and decoding parameters. We report rollout diagnostics and approximate text-to-motion metrics. Rollout diagnostics measure duration, early stopping, root-path coverage, motion velocity, and repetition. For text-to-motion metrics, generated kinematic motions are adapted to approximate HumanML3D 22-joint, 263-dimensional features and evaluated with a pretrained HumanML3D text-motion evaluator. We report FID, R-precision, and matching score; FID and matching are lower-is-better, while R-precision is higher-is-better.

The selected fine-tuned model improves R@2, matching score, average generated length, early-stop rate, and root-path coverage on Long100 (Table III). It ties R@1 and R@3, while FID is slightly worse. These results indicate that the fine-tuned model improves several long-prompt rollout and retrieval indicators, while distribution-level motion quality as measured by the approximate FID does not improve in this setting.

Qualitative video inspection follows the same trend. Compared with the released baseline, the fine-tuned model shows stronger long-horizon motion generation: it more often main-

TABLE IV
STAGE II REFERENCE PREPROCESSING DIAGNOSTICS. ROOT HEIGHTS ARE IN METERS; JOINT VELOCITY AND ACCELERATION ARE MEAN JOINT-SPACE NORMS, WHERE LOWER IS BETTER.

Motion set	Root ₀ z	Min root z	$\ \dot{q}\ \downarrow$	$\ \ddot{q}\ \downarrow$
Baseline	0.969	0.206	6.542	169.627
Short raw	0.591	0.435	33.739	547.296
Short processed	1.058	0.694	15.820	246.378
Long raw	0.533	0.206	3.013	35.961
Long processed	1.059	0.732	2.982	35.299

TABLE V
STAGE II RETARGETING ACCELERATION RESULTS.

Setting	Time	Speedup
Original H1 fitting	72 h	1.00×
Optimized fitting	68 h	1.06×
Batched conversion	3.2 h	22.5×

tains motion over the requested horizon, reduces immediate early stopping, and covers longer root paths on representative long prompts. Representative Stage I comparison videos are provided on the project page: <https://aeslitech.github.io/L-PHYM/>.

C. Stage II: Retargeting and Teacher Tracking

Stage II evaluates whether the kinematic references can be converted into physically meaningful H1 motion libraries and tracked by a HOVER-style teacher policy. We first diagnose the reference motions before policy rollout. This is necessary because a motion that is visually plausible in a skeleton viewer can still initialize H1 at an invalid height or demand unrealistically sharp joint changes. Table IV compares the original HOVER reference set with raw and processed generated references. Preprocessing lifts the generated references into a plausible standing-height range and reduces joint-space smoothness outliers, especially for the short generated reference.

The retargeting bridge is also optimized for batched generated references. Table V reports the wall-clock cost of the H1 fitting and conversion pipeline. Minor optimizer-level changes reduce runtime only slightly, while batched conversion with cache reuse makes large-scale reference preparation practical for the final experiments.

We then evaluate the trained teacher checkpoint on motion sets from the H1 AMASS-style benchmark directory [? ? ? ? ? ?]. All runs use the same teacher, the same evaluation script, and ten parallel Isaac Lab environments. Table VI reports several motion sets found in this screening pass. The teacher tracks compact and well-grounded references most reliably, with the strongest results on DFaust, SSM, and MoSh.

V. ANALYSIS

A. Stage I: Hand-Coded State Mapping

The first complete Stage I attempt used the direct HumanML3D-to-MoConVQ state route. It composed Hu-

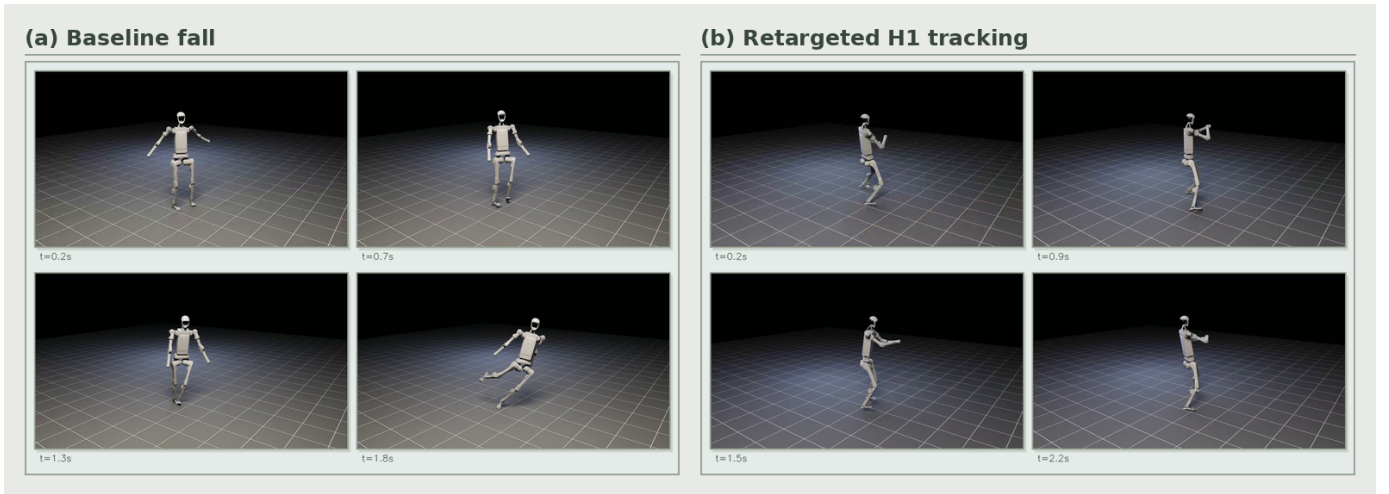


Fig. 3. Stage II visual comparison. The baseline rollout loses balance and falls, while the retargeted H1 tracking example remains upright over the sampled sequence. These frames illustrate the role of Stage II in turning generated kinematic references into physically tracked humanoid motion.

TABLE VI

TEACHER TRACKING BENCHMARKS FROM H1 MOTION SETS. SUCCESS IS HIGHER-IS-BETTER; MPJPE METRICS ARE COMPUTED OVER ALL ROLLOUTS IN MILLIMETERS AND ARE LOWER-IS-BETTER.

Benchmark	Motions	Success	Succ. rate	MPJPE-G	MPJPE-L	MPJPE-PA
DFaust	129	117/129	90.70%	129.14	84.73	67.25
SSM	30	23/30	76.67%	115.33	81.11	69.14
MoSh	77	56/77	72.73%	121.88	86.07	70.28
HumanEva	28	17/28	60.71%	128.24	84.42	69.27
ACCAD	252	109/252	43.25%	141.91	91.84	72.60
HUMAN4D	148	58/148	39.19%	138.93	91.75	74.72

manML3D long sequences, converted them to MoConVQ body states with hand-coded mapping rules, encoded the resulting observations into RVQ tokens, and fine-tuned the text-conditioned GPT. The training curves looked promising: a later checkpoint reached 0.5569 training token accuracy and 0.5236 validation token accuracy. However, generated videos were unstable, repetitive, and weakly aligned with the prompts.

Figure 4 illustrates why this route was misleading. Loss and token accuracy improved, but token-distribution and observation diagnostics showed that the cache was off distribution for MoConVQ, with residual-VQ token collapse. This study rejected the hand-coded state route and established the main diagnostic lesson: token accuracy is not a sufficient success criterion when the data-to-token interface is wrong.

B. Stage I: Synthetic-Data Scaling

The full1000 run starts from 1000 synthesized long sequences. After native retarget diagnostics and quality filtering, 403 sequences are accepted and split into 322 training and 81 validation sequences. The larger cache is much easier to fit: validation loss reaches 6.3994 and validation token accuracy reaches 0.0821 after three epochs. This result confirms that the larger filtered cache provides a stronger optimization signal than the selected small cache.

This optimization gain does not transfer to approximate paper-style metrics (Table VII). The reason is that the filter

TABLE VII

FULL1000 SCALING RUN ON LONG100. IT IMPROVES OPTIMIZATION AND SOME ROLLOUT DIAGNOSTICS, BUT NOT APPROXIMATE FID/R-PRECISION.

Metric	Baseline	Epoch 2	Epoch 3
Avg. frames	1216.56	1200.96	1200.24
Early-stop rate	0.44	0.41	0.41
Root path	3.3679	3.4410	3.3185
Lag20 repeat frac.	0.0120	0.0059	0.0035
Approx. FID	7.0400	8.3016	7.9617
Approx. R@1	0.050	0.050	0.050
Approx. R@2	0.110	0.090	0.110
Approx. R@3	0.160	0.110	0.160
Approx. matching	5.0077	5.0335	5.0802

removes obvious retarget and token outliers but does not ensure semantic compatibility, caption granularity, or useful boundary supervision. The larger accepted set is therefore more heterogeneous and can pull the fine-tuned model away from the pretrained distribution rewarded by the HumanML3D evaluator. The result shows that the next bottleneck is data quality and semantic filtering, not only the number of accepted native-retargeted samples.

C. Stage I: Approximate Evaluation Protocol

The Long100 metrics use an approximate evaluation protocol because Stage I generates native-retargeted kinematic trajectories rather than motions stored in the original HumanML3D representation. Before applying the pretrained HumanML3D text-motion evaluator, each generated trajectory is converted to approximate 22-joint, 263-dimensional HumanML3D features. The reported FID, R-precision, and matching score therefore measure performance after this adapter, rather than a native SMPL/HumanML3D benchmark evaluation.

This protocol is still useful for model comparison because the released baseline and the fine-tuned model share the same prompts, decoding settings, feature adapter, and evaluator.

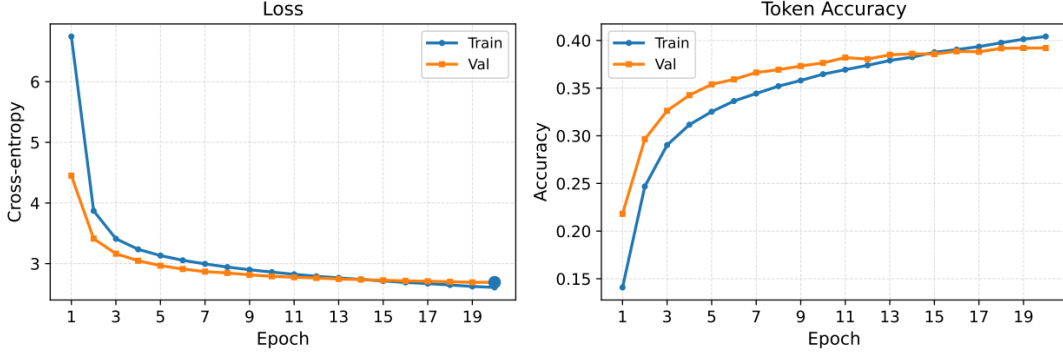


Fig. 4. Exploratory hand-coded mapping run. Loss and token accuracy improve, but qualitative rollouts remain poor, showing that cache fit alone is not a reliable success criterion when the HumanML3D-to-MoConVQ representation mapping is off distribution.

However, the values should be interpreted as relative Stage I evidence under a fixed adapter-based protocol. Long100 is also a text-only held-out prompt suite, so it measures generation from unseen multi-segment instructions rather than reconstruction against a smoothed ground-truth long-motion corpus.

D. Stage II: Retargeting Diagnostics

Stage II repeats the main lesson from Stage I at a physical-control interface. In Stage I, visually reasonable motion was not enough if the representation fed to MoConVQ was off distribution. In Stage II, visually reasonable skeleton motion is not enough if the H1 reference starts at an invalid height, contains sharp joint changes, or creates contact patterns that the teacher cannot track. Table IV shows this directly. The short raw reference has much larger joint velocity and acceleration norms than the baseline; after processing, both quantities are reduced substantially and the root height is lifted into a plausible H1 range. The long reference is already smoother in joint space, but still needs height normalization before it becomes a reliable policy-training reference.

The speed study in Table V identifies a separate systems bottleneck. Optimizer-level changes only reduce the H1 fitting time from 72 hours to 68 hours, so the objective itself is not where most practical time is saved. Batched conversion and cache reuse reduce the conversion stage to 3.2 hours, making broad screening over multiple motion sets feasible. This matters for the report because Stage II evidence depends on evaluating several candidate reference libraries, not only a single hand-picked clip.

E. Stage II: Teacher Tracking Screening

The teacher benchmark is best interpreted as a screening experiment rather than a completed deployment result. Every row in Table VI uses the same checkpoint, evaluation script, and environment count, so the ranking mainly reflects motion-set compatibility with the current retargeting and teacher-training distribution. DFaust, SSM, and MoSh give the clearest positive evidence: the teacher completes most motions in these sets while maintaining all-rollout MPJPE in the same range as the lower-success rows. HumanEva remains a moderately

successful transfer case, while ACCAD and HUMAN4D show that success drops once the screening set becomes larger and more diverse.

The most informative signal is therefore the success rate, not small differences between MPJPE values. MPJPE-G and MPJPE-PA remain relatively clustered across the selected rows, while completion rate drops from 90.70% on DFaust to 39.19% on HUMAN4D. This pattern suggests that the dominant Stage II failure mode is whole-episode feasibility and stability, rather than only per-frame pose matching after a stable rollout has been achieved. In other words, when the teacher remains upright and synchronized with the reference, its pose-tracking quality is fairly consistent; the main gap is whether it can keep the rollout alive across the full motion.

Overall, the current teacher policy is effective on compact, well-grounded motion libraries and already provides strong evidence that the retargeted H1 references are physically trackable. Its weaker results on ACCAD and HUMAN4D indicate sensitivity to broader motion diversity, longer effective horizons, and harder contact or balance transitions. The next improvement should therefore focus on strengthening the teacher itself through better reference filtering, contact-aware retargeting checks, and curriculum sampling over the motions that produce early termination or loss of balance.

VI. CONCLUSION

L-PHYM connects long-horizon language-conditioned motion generation with physics-based H1 tracking. Stage I uses boundary-aware HumanML3D compositions and MoConVQ-native fine-tuning to avoid the off-distribution token cache caused by direct hand-coded state conversion, improving several Long100 rollout and retrieval indicators while leaving approximate FID slightly worse. Stage II retargets generated references to H1 and shows that compact motion sets can be tracked by a HOVER-style teacher, while broader sets expose remaining whole-motion feasibility issues in balance, contact consistency, and synchronization. Future work should improve semantic filtering, duration modeling, native HumanML3D/SMPL evaluation, retarget-quality checks, and long-horizon tracking curricula.

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